

FieldBox

A rugged, self-contained **Tailscale subnet router** in a Pelicase. Drop it into a venue or production network and the rest of your tailnet reaches that LAN — over wired internet or a 4G prepaid modem. Built to be run by a non-technical operator: power on → green light → done.

2× Ethernet + 4G auto-failover

Pre-locked to your tailnet

Auto-detect subnet

Self-healing (watchdog)

Touchscreen UI

This pack finalises every open decision from the plan (display bus, carrier board, 4G interface, failover metrics, route-approval detection), plus schematics, a case mockup, and a screenshot of the working dashboard. Repo: github.com/jimdolman/fieldbox

Prepared for Jim Dolman · 16 July 2026 · Difficulty: moderate — bench prototype ~1 day, polished cased unit ~1 week part-time. Parts ~€330–420/unit.

1 What it is & how hard it is

The subnet-routing itself is a solved, robust problem — a couple of Tailscale commands plus one ACL rule, and Tailscale's default masquerade means the venue network needs *zero* changes. The real work is three things: the idiot-proof touchscreen GUI, the wired↔4G failover, and the Pelicase fabrication. None are hard; the time simply goes into polish and the case.

AREA	EFFORT	NOTES
Tailscale subnet router	Easy	Well-trodden; masquerade removes all venue-side config.
Dual-WAN + 4G failover	Easy–moderate	NetworkManager route metrics + connectivity check.
Touchscreen GUI	Moderate	Already scaffolded & running (see §7). Mostly polish.
Pelicase fabrication	Moderate	Panel-mount cutouts, screen in lid, cooling — the manual bit.

2 Finalised decisions (every open item resolved)

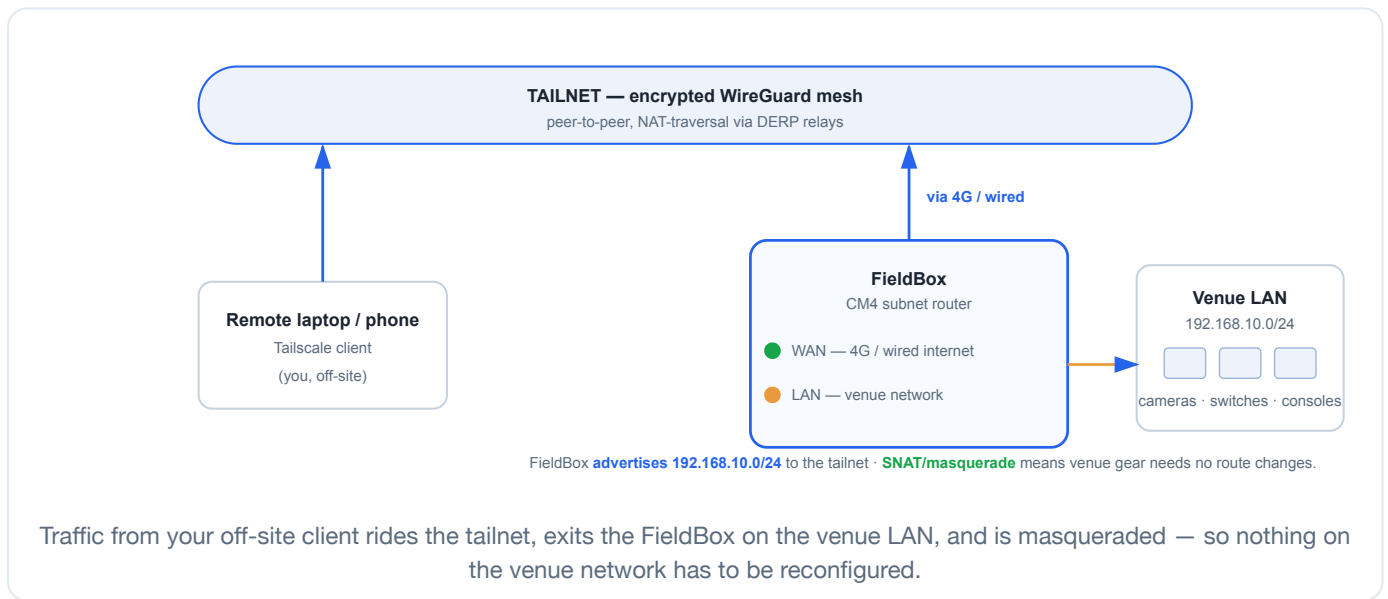
OPEN ITEM	STATUS	DECISION & REASON
Display bus (DSI vs HDMI)	FIXED	HDMI. The chosen dual-GbE CM4 carrier exposes only micro-HDMI (no DSI header), so use a 5" HDMI capacitive touchscreen (800×480) via a short micro-HDMI ribbon + USB for touch.
Carrier board	FIXED	Seeed Dual-GbE CM4 carrier (reRouter CM4001). Two <i>true</i> GbE (native + PCIe RTL8111), 2× USB 3.0, micro-HDMI, microSD, USB-C power. Real NICs beat a USB dongle for stability.
Compute module	FIXED	CM4104032 — 4 GB RAM, 32 GB eMMC, Wi-Fi/BT. eMMC avoids SD-card failures; Wi-Fi is a free third WAN later.
4G modem & interface	FIXED	Huawei E3372h-320 (HiLink/ECM). Appears as a DHCP Ethernet interface — no ModemManager/PPP. Pinned to <code>wwan0</code> , route metric 700. Prepaid SIM, PIN disabled .
WAN failover logic	FIXED	NetworkManager metrics: <code>wan0 = 100</code> (preferred), <code>wwan0 = 700</code> (backup), <code>lan0 = never-default</code> . NM's connectivity check demotes a dead link automatically; re-plugging fails back.
Route-approval readout	FIXED	Dashboard reads <code>Self.PrimaryRoutes</code> from <code>tailscale status --json</code> (routes the node actively serves = approved). Advertised but not present ⇒ shows “pending”. (Prior TODO removed.)
accept-routes	FIXED	Kept <code>--accept-routes=false</code> (pure one-way subnet router). Documented how to flip on if the box ever needs to reach <i>other</i> tailnet subnets.
UI reboot privilege	FIXED	Locked-down sudoers: kiosk user may run only <code>/sbin/reboot</code> (<code>NOPASSWD</code>), nothing else.
Onboarding	FIXED	Pre-locked to your tailnet via a tagged, reusable auth key + ACL <code>autoApprovers</code> for RFC1918. Key expiry disabled on the device. No login, ever.
Power / autonomy	FIXED	Mains USB-C, no battery . Hardware watchdog + 1-min health timer self-heal instead (“if the site's down there's nothing to route”).

3 Bill of materials

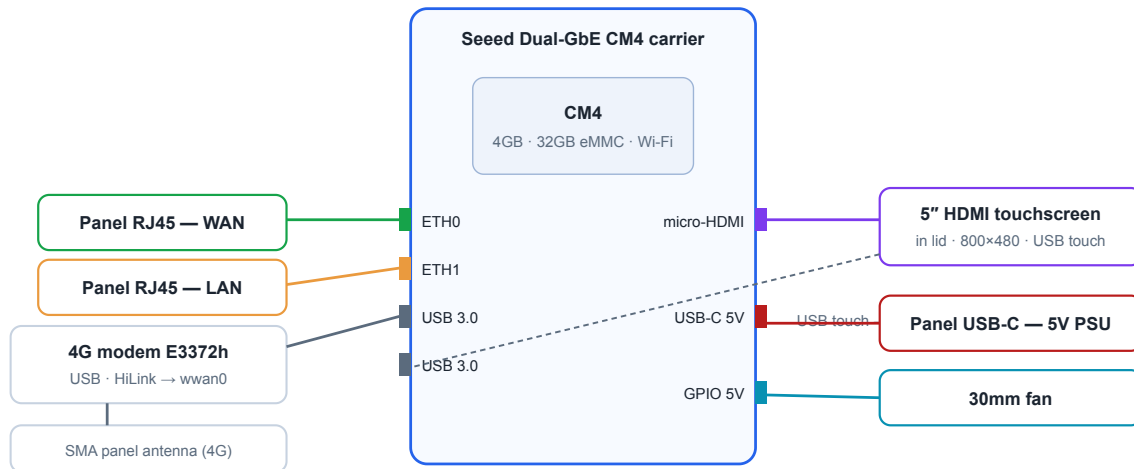
#	PART	CHOSEN ITEM	EST. €
1	Compute module	Raspberry Pi CM4104032 (4GB/32GB/Wi-Fi)	75
2	Carrier board	Seeed Dual Gigabit Ethernet CM4 carrier (reRouter)	60
3	4G modem	Huawei E3372h-320 (HiLink) + prepaid data SIM	45
4	Touchscreen	5" HDMI capacitive, 800×480 (Waveshare)	50
5	PSU	Official 5V/3A USB-C	12
6	Cooling	Heatsink + 30mm 5V fan + Gore pressure vent	18
7	Case	Pelicans 1150/1200	55
8	Panel & cabling	2× RJ45 feedthrough, SMA bulkhead + whip, USB-C panel mount, standoffs, short patch/HDMI/USB leads	45
Total		per unit (ex. SIM plan)	~360

Second and subsequent units are flash-and-go from the golden image; only hardware cost repeats.

4 How it works — data flow



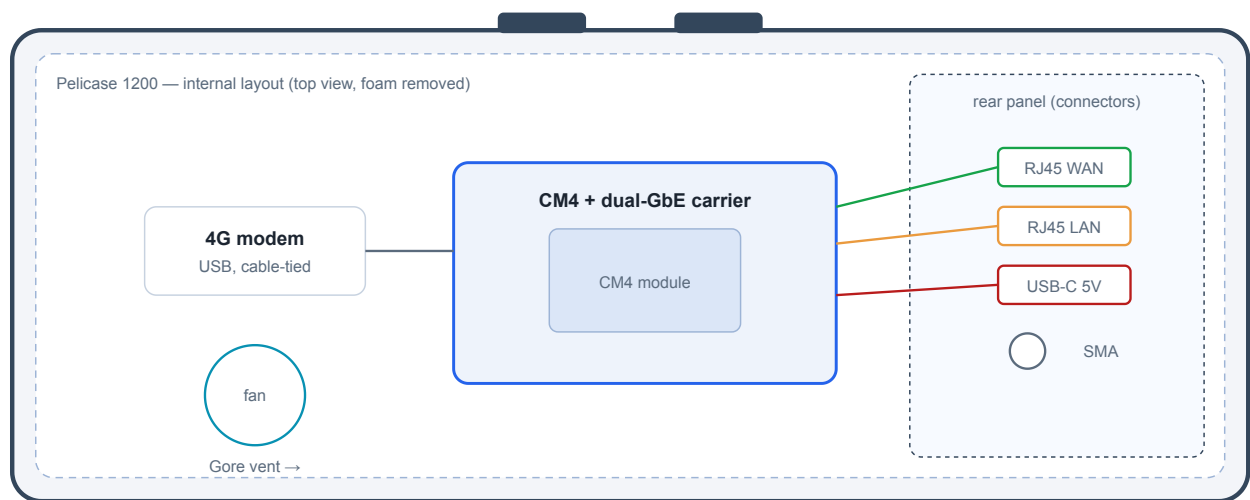
5 Internal wiring / block diagram



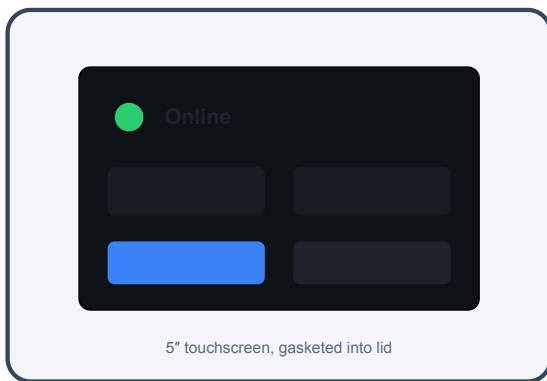
Colour key: green = WAN Ethernet, amber = LAN Ethernet, grey = USB, violet = display, red = power, cyan = fan.

Interface names are pinned by MAC in `/etc/systemd/network/*.link` so `wan0` / `lan0` / `wwan0` never swap around between boots.

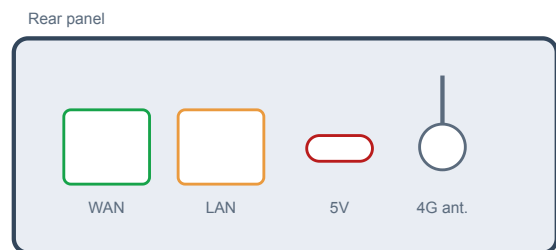
6 Case mockup



Top-down internal arrangement. The touchscreen mounts in the *lid* (see panel views below).



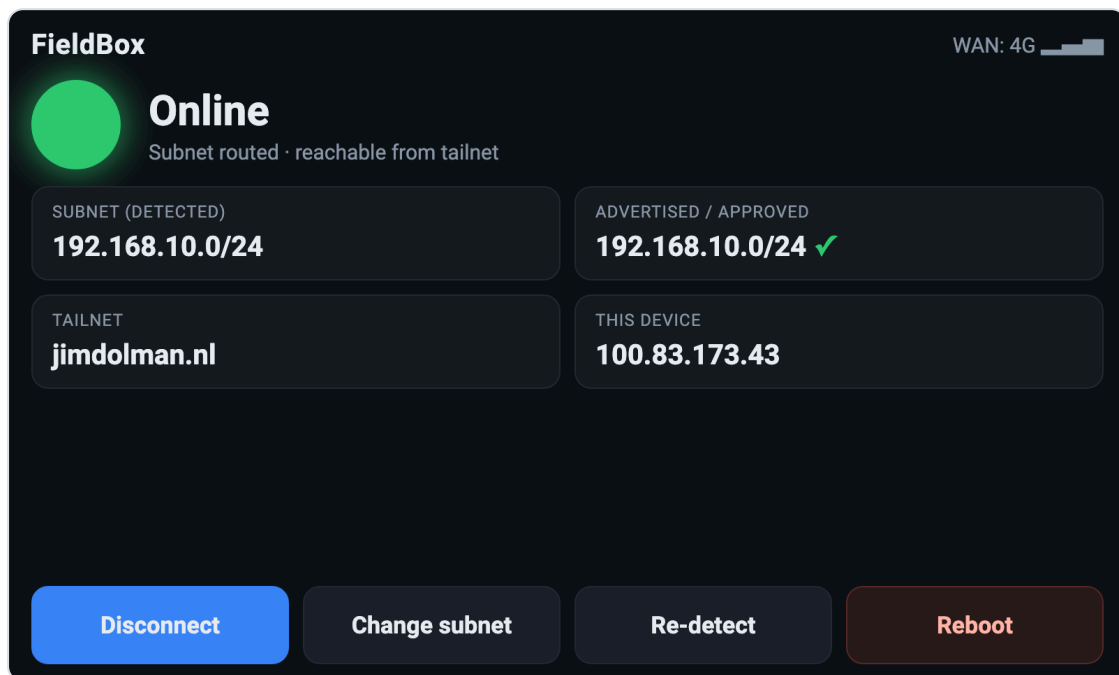
Lid — operator view



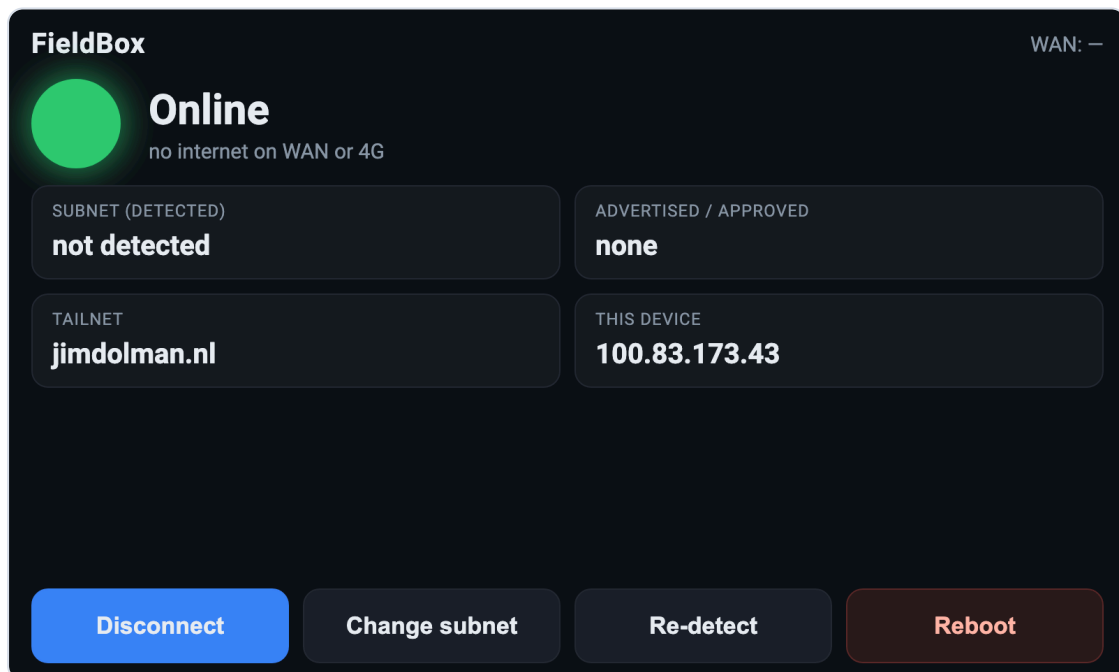
Rear panel — colour-coded, labelled ports

7 Touchscreen dashboard

The GUI is a local FastAPI app shown fullscreen by a `cage` + Chromium kiosk. It's already built and running in the repo. Below: a representative *field* state (simulated for the shot), then the actual live app running on the dev machine.



Representative field state (simulated): 4G WAN, subnet `192.168.10.0/24` advertised & approved (✓).



Live, unmodified app rendering real Tailscale data on the dev Mac (no `lan0` / `wan0` there, so subnet/WAN read empty — exactly the graceful off-device behaviour).

Operator flow: power on → box auto-joins the tailnet → auto-detects & advertises the LAN subnet → light goes green. The only control they ever need is **Change subnet** / **Re-detect**.

8 Software & provisioning

COMPONENT	ROLE
<code>fieldbox-ui.service</code>	FastAPI backend serving the dashboard + JSON API (status, set-subnet, connect, reboot).
<code>fieldbox-kiosk.service</code>	<code>cage</code> + Chromium fullscreen on the touchscreen.
<code>fieldbox-firstboot.service</code>	One-shot <code>tailscale up</code> with the stored tagged auth key; advertises the detected subnet.
<code>fieldbox-health.timer</code>	Every 60 s: verify tailscaled + WAN + backend; restart / renew / (last resort) reboot.
HW watchdog	BCM2711 watchdog via systemd — auto-reboot on a hard hang.
<code>tailscale/acl.hujson</code>	<code>tagOwners</code> + <code>autoApprovers</code> for RFC1918 → any detected subnet auto-approves, no admin click.

Per-unit setup (one-time)

- Flash Pi OS (Bookworm) → run `scripts/setup.sh`.
- Set the two MACs in `network/*.link` (find with `ip -br link`), reboot.
- Drop a tagged, reusable auth key into `/var/lib/fieldbox/authkey` (`chmod 600`).
- Paste `acl.hujson` blocks into tailnet ACLs; disable key expiry for the device.

9 Verification & test checklist

TEST	PASS CRITERIA
Boot & naming	<code>ip -br link</code> shows <code>wan0</code> , <code>lan0</code> , <code>wwan0</code> .
WAN failover	Wired up → default via <code>wan0</code> . Unplug → 4G takes over within seconds, tailnet stays up. Re-plug → fails back.
Subnet routing	From an off-site client: route shows approved ; <code>ping</code> + open a venue device's web UI works with <i>no</i> change on that device.
Subnet change	Move <code>lan0</code> to a new subnet → Re-detect → new CIDR advertised, auto-approved, reachable.
Self-heal	<code>kill -9 tailscaled</code> → health timer restarts it. Simulate hang → board reboots and returns to Online unattended.
Idiot test	Cold power-cycle → within ~60–90 s the screen shows Online + detected subnet with zero interaction.

Buy-early checks: confirm the exact carrier revision's display output (micro-HDMI) before ordering the screen; confirm the E3372h variant is the *HiLink* (usb-ethernet) firmware, not the stick/PPP one; disable the SIM PIN before first insert.

10 Shopping list — everything to buy (one unit)

Ballpark EUR, ex. shipping and SIM plan. Full checklist also in [docs/SHOPPING-LIST.md](#).

GROUP	ITEM	QTY	EST. €
Core	Raspberry Pi CM4104032 (4GB/32GB/Wi-Fi)	1	75
	Seeed Dual-GbE CM4 carrier (reRouter CM4001)	1	60
	Huawei E3372h-320 4G modem (HiLink)	1	35
	Prepaid data SIM (mini / 2FF)	1	plan
	5" HDMI capacitive touchscreen 800×480	1	45
Power	Official 5V/3A USB-C PSU	1	12
	Panel-mount USB-C feedthrough	1	8
Net & RF	RJ45 Cat6 feedthrough couplers (WAN + LAN)	2	10
	Short Cat6 patch leads ~0.25 m	2	6
	TS-9 → SMA pigtail	1	6
	SMA 4G whip antenna	1	8
Display	micro-HDMI → HDMI cable ~0.3 m	1	6
	USB-A → micro-USB cable (touch)	1	3
Cooling	CM4 heatsink	1	5
	30 mm 5V fan	1	4
	Gore pressure-equalization vent	1	8
Case & mech	Pelicase 1150 / 1200	1	55
	M2.5 standoff + screw kit	1	8
	Cable ties + adhesive mounts	1	4
	Connector panel (blank / 3D-printed)	1	5
	Thermal pads / paste	1	3
Flashing	USB-A ↔ USB-C data cable (rpiboot) — may own	1	4
Total (ex. shipping + SIM plan)			~370

Confirm before ordering: (1) the E3372h is the **HiLink** variant, not stick/PPP; (2) the carrier revision's display output is **micro-HDMI** (matches the screen); (3) disable the SIM PIN and disable Tailscale key expiry.